

Louay Yahyaoui

Machine Learning Engineer

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Summary

ML engineer with hands-on experience across the model lifecycle, from research and fine-tuning to production optimization and deployment on edge devices. Comfortable working across CV, NLP, and speech domains.

Work Experience

- **AI Engineer Intern (End-of-Studies Internship)** Mar 2025 - Sep 2025
EssilorLuxottica *Sophia-Antipolis, France*
 - Enabled real-time speech-enhancement for edge devices by optimizing production models, reducing its size and activation footprint by 75% with under 1% accuracy loss using PTQ and QAT.
 - Built a reusable PyTorch and AIMET → ONNX export pipeline, adapted to the hardware team's requirements, enabling direct integration into the target SoC.**Tools:** PyTorch, ONNX, Qualcomm AIMET, OVHCloud, Git.
- **AI Research Intern** Jun 2024 - Aug 2024
MUST University *Tunis, Tunisia*
 - Reviewed research on AI applications in education and proposed an AI assistant to improve the university student experience, setting the technical direction for the project.
 - Built the assistant as an academic Q&A proof-of-concept by generating a GPT-4 synthetic dataset, fine-tuning Llama 3.1 (8B) with QLoRA, and integrating the model into a functional demo.**Tools:** PyTorch, Hugging Face Transformers, Weights & Biases.
- **Data Science Intern** Jun 2023 - Jul 2023
Hexastack *Tunis, Tunisia*
 - Developed an RNN language classifier for chatbot routing, reaching 99.5% accuracy on a small internal held-out set.
 - Updated the training and validation workflow, end-to-end, for an existing BERT intent detection and slot-filling pipeline, improving intent accuracy to 99.8% and NER accuracy to 97%.**Tools:** TensorFlow, Pandas, NumPy, Git.

Projects

- **Deepfake Video Detector** - Reproduced the GenD paper by implementing a CLIP-based deepfake detector, fine-tuning only LayerNorm parameters (0.03% of weights), achieving 85.9% AUROC on DFDC.
- **Lightweight GPT-style Language Model** - Implemented a GPT-style language model on the TinyStories dataset (2.7M short stories) for end-to-end next-token prediction, yielding a val perplexity of 3.9.
- **Satellite Image Segmentation** - Trained a TensorFlow U-Net to segment complex geospatial structures on the DeepGlobe land cover dataset, attaining 81% IoU with strong spatial consistency.
- **AskDocs - RAG Pipeline over arXiv Papers** - Built a retrieval-augmented QA system with LangChain: ingests papers via the arXiv API, chunks and embeds their text into a Qdrant vector store, then generates context-grounded answers with an open-source LLM (Qwen2.5-7B).

Education

- **Engineering Diploma in Software Engineering** Sep 2020 - Oct 2025 (Graduated)
National Institute of Applied Science and Technology (INSAT)

Skills

- **ML/DL:** PyTorch, TensorFlow, Hugging Face, QLoRA, LangChain, Qdrant, prompt engineering
- **Model Optimization & Deployment:** ONNX, Qualcomm AIMET, PTQ/QAT
- **Software & Tooling:** Python, C++, TypeScript, SQL, Bash, Linux, Git, CI/CD, Docker
- **Cloud & Infra:** OVHCloud, Azure
- **Languages:** English (Fluent), French (Advanced), Arabic (Native), Spanish (Elementary)

Certifications

- **DeepLearning.AI:** Machine Learning, Deep Learning, NLP Specializations (Coursera 2024)